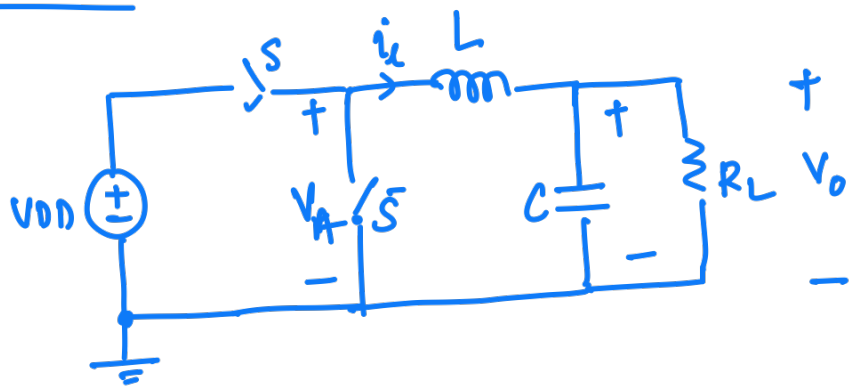
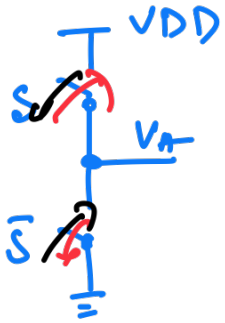
Lecture #24







$V_o = D \cdot VDD$

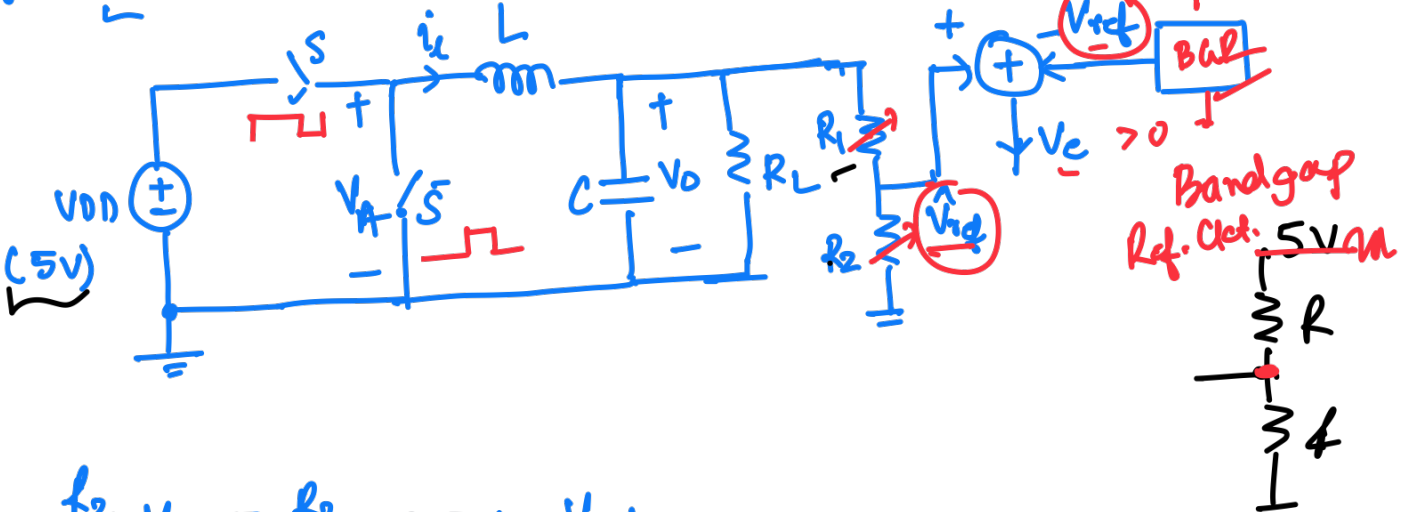
$\Delta i_L = \frac{VDD}{L} \frac{1}{f_s} D(1-D)$; $D = \frac{T_H}{T}$

$\Delta V_c = \frac{VDD}{8} \frac{1}{LC} \frac{1}{f_s^2} D(1-D)$

$= \frac{VDD}{8} \frac{\omega_p^2}{f_s^2} D(1-D)$; $\omega_p = \frac{1}{\sqrt{LC}}$

- Increase f_s to reduce the ripple.
- Switching losses increases with f_s.
- Power efficiency ~ 95% - 99%

V_o = 2.5, 1.8, 1.2, 1.0, 0.9 ..



$\hat{V}_{ref} = \frac{R_2}{R_1+R_2} V_o = \frac{R_2}{R_1+R_2} \times 2.5V = V_{ref}$ (fix)

$V_{des} = 2.5V \lessgtr V_o$

$V_o \times \frac{R_2}{R_1+R_2} \lessgtr V_{ref}$ fix

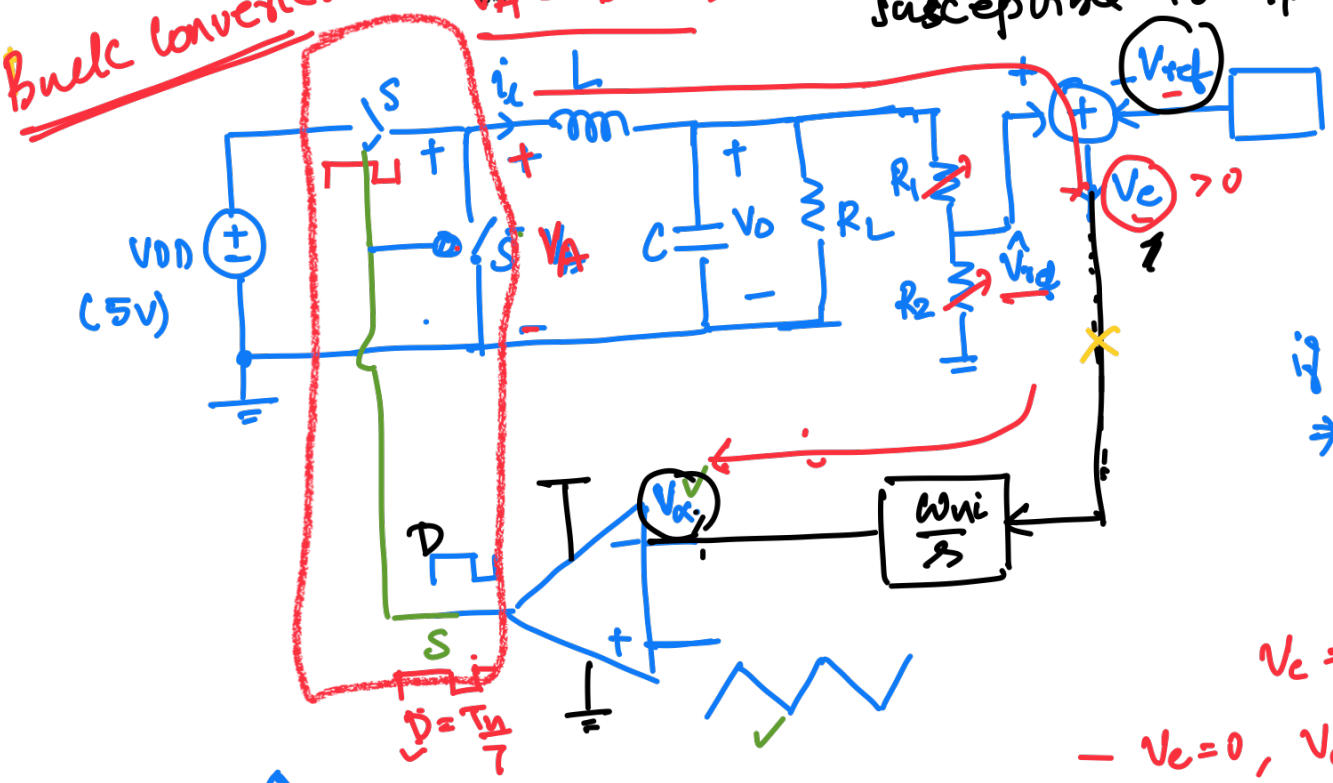
I want 2.5V supply.

I want 2.5V supply to provide large current to the load.

I want my output supply least susceptible to i/p supply vol.

"Buck converter"

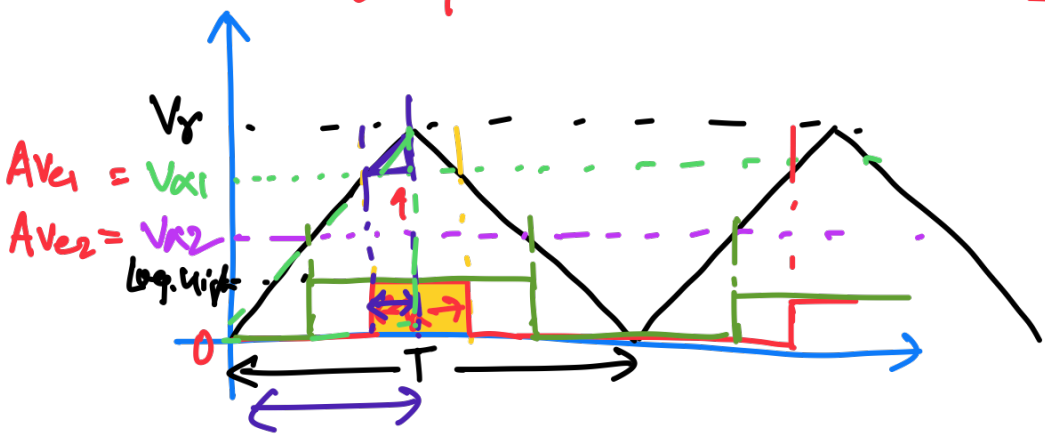
$V_A = D \cdot V_{DD}$

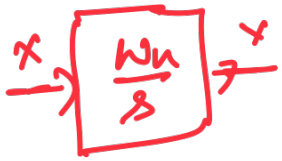


if $\hat{V}_{fb} > V_{ref}$
$\Rightarrow V_e > 0$

$V_e = 0 \Rightarrow A_{Ve} = 0$

– $V_e = 0$, $V_\alpha = f(V_e)$ = non-zero





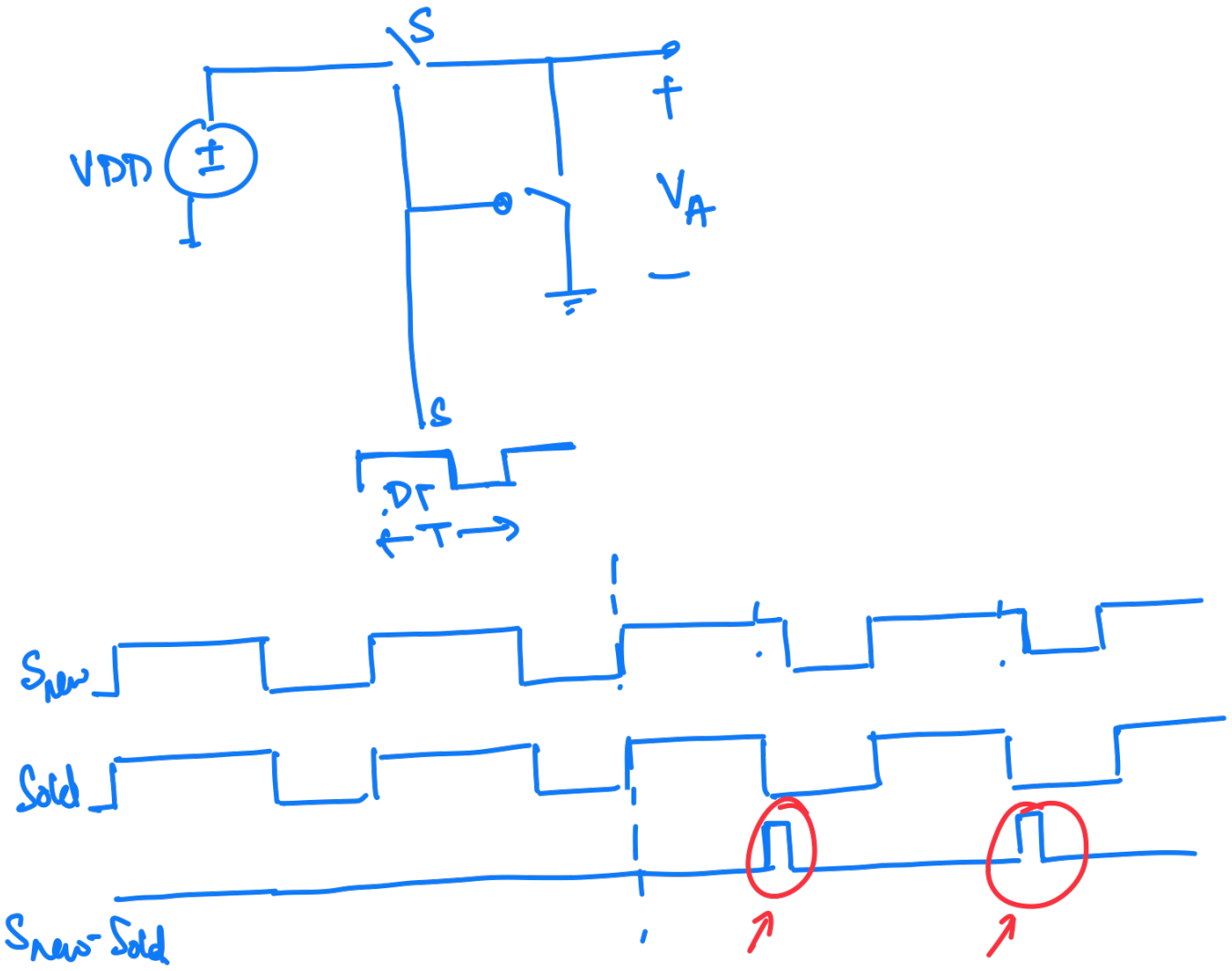
$\frac{y}{x} = \frac{\omega_n}{s}$

$\left.\frac{y}{x}\right|_{s=0} = \infty$

$$V_\alpha(s) = \frac{\omega_{ni}}{s} V_e(s)$$

$$\frac{V_\gamma}{T/2} = \frac{(V_\gamma - V_\alpha)}{T_H/2} \Rightarrow \frac{T_H}{T} = \left(1 - \frac{V_\alpha}{V_\gamma}\right)$$

$$\boxed{D = 1 - \frac{V_\alpha}{V_\gamma}}$$


S
VDD
+
VA
−
s
DT
T
Snew
Sold
Snew-Sold